

WHAT WE FOUND

The bottleneck helps **only** when labels are scarce.
Give it enough labels and **plain hierarchical MLM wins**.

N = 10

GlocalDoc **0.044** vs H-MLM 0.019 → **2.3×**, bottleneck wins

N $\geq$ 50

H-MLM **0.273** vs GlocalDoc 0.108 → **the ordering flips**

1 Why & what we ask

Why.

- ▶ Language models capture *local* meaning but lose the *whole* document; in a legal judgment the decisive signal is spread across many paragraphs.
- ▶ Labelling ECtHR cases needs expert effort, so the realistic setting is **few-shot**.

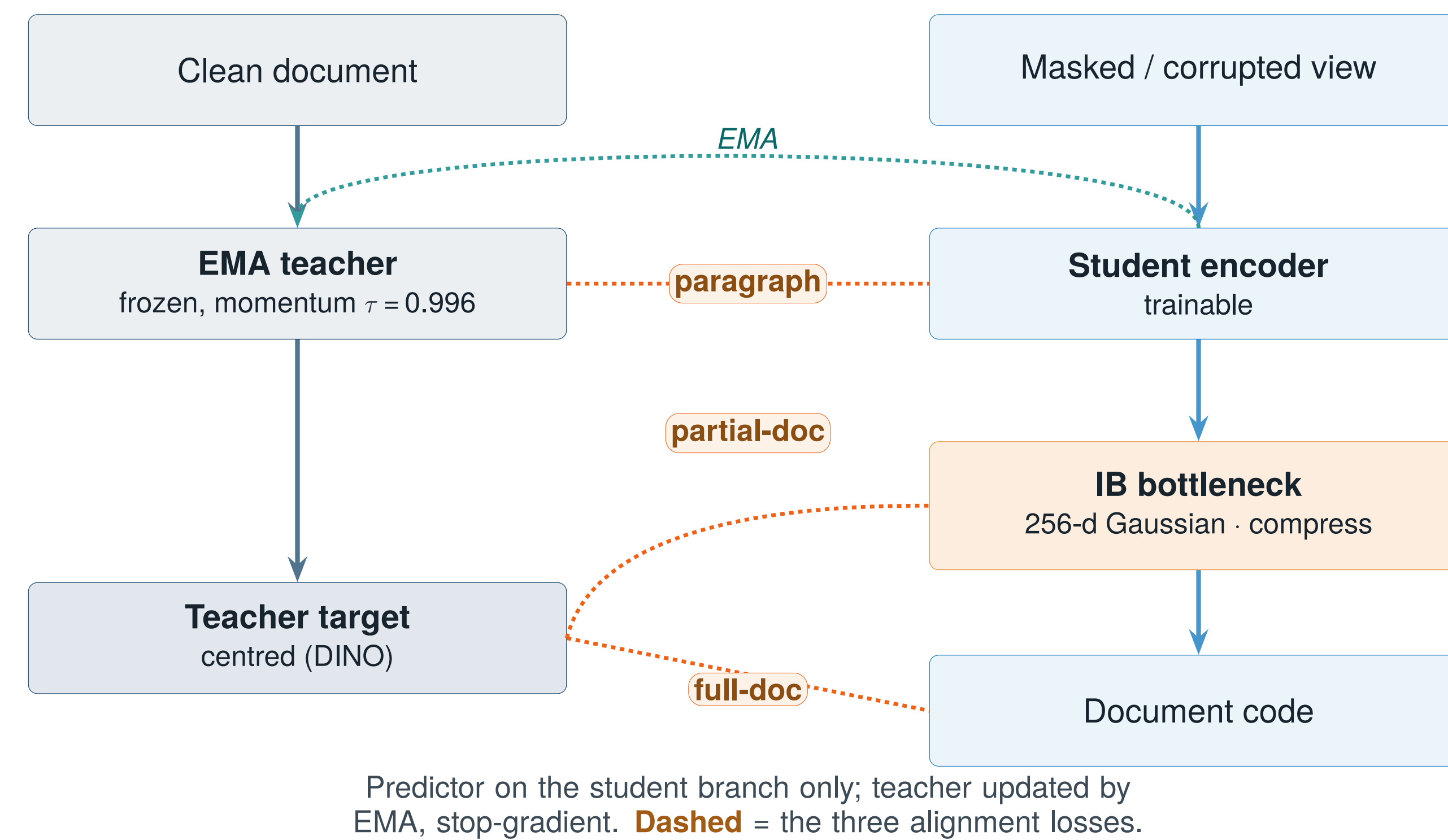
We ask.

- ▶ **RQ1** Beat hierarchical MLM for few-shot document features?
- ▶ **RQ2** Does the advantage depend on the number of labels N ?
- ▶ **RQ3** What breaks on heavy boilerplate, and what fixes it?

DATA

ECtHR judgments (LexGLUE), **10** article labels.
5 to 50 paragraphs per case, about **100×** class imbalance.
 Primary metric: **macro-F1**.

2 How GlocalDoc works



5 Results

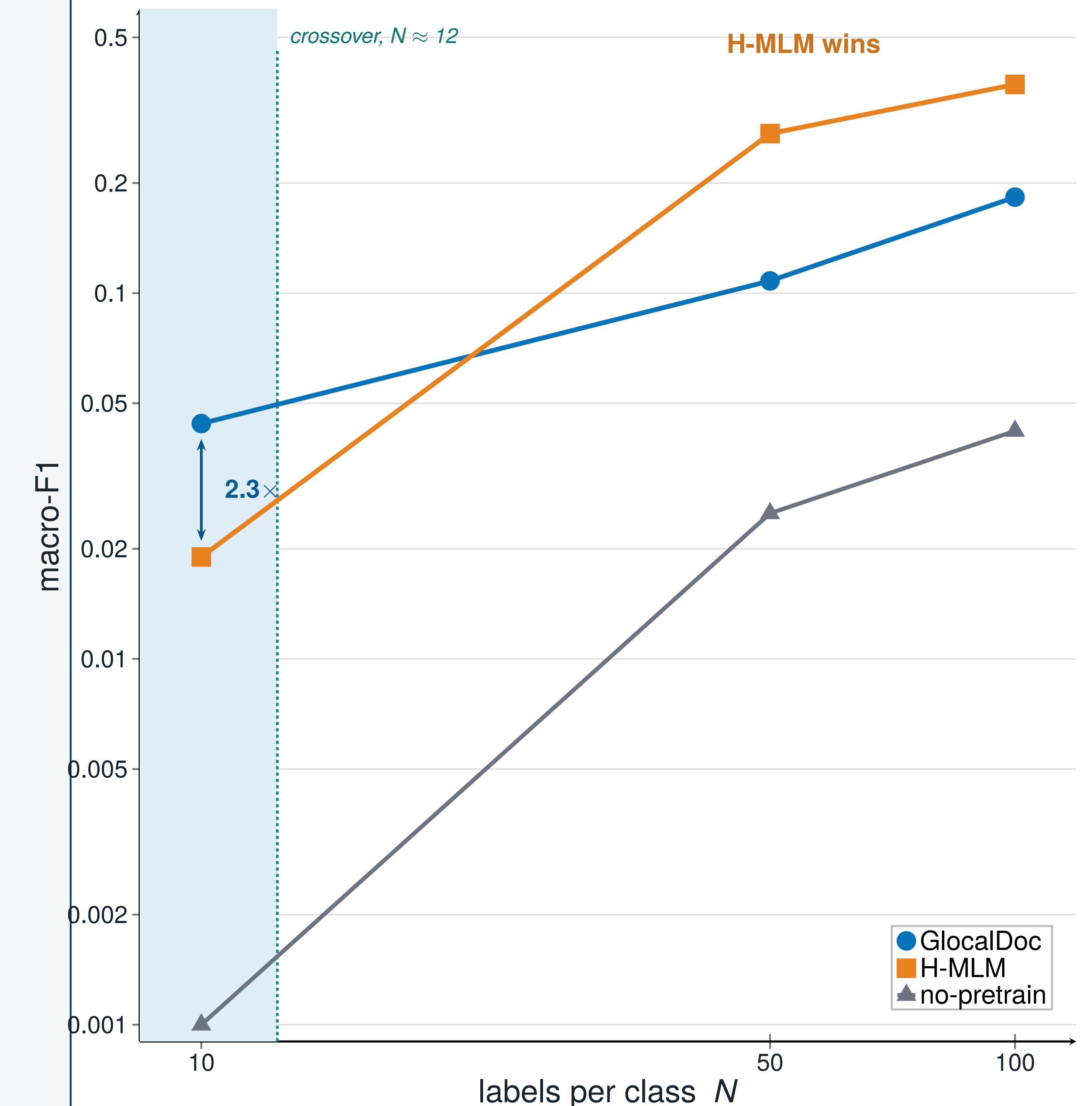
Macro-F1, mean $\pm$ sd over 5 seeds

Pre-training	N=10	N=50	N=100
GlocalDoc	0.044 $\pm$.009	0.108 $\pm$.024	0.183 $\pm$.006
H-MLM	0.019 $\pm$.023	0.273 $\pm$.061	0.372 $\pm$.032
no-pretrain	0.001	0.025	0.042

Winner per column in **bold**. At N=10, **2 of 5** H-MLM seeds collapse to ≈ 0 while all 5 GlocalDoc seeds stay in **0.034 to 0.057**: stability, not just a higher mean.

3 The crossover

Macro-F1 vs labels per class N , **log scale** (5 seeds). Glocal-Doc wins at $N = 10$; H-MLM takes over once labels are plentiful.



Bottom line: the bottleneck is a strong prior. A lifesaver under extreme scarcity, a handicap once labels are plentiful (it discards the lexical detail the MLM keeps). **Match the objective to the data regime.**